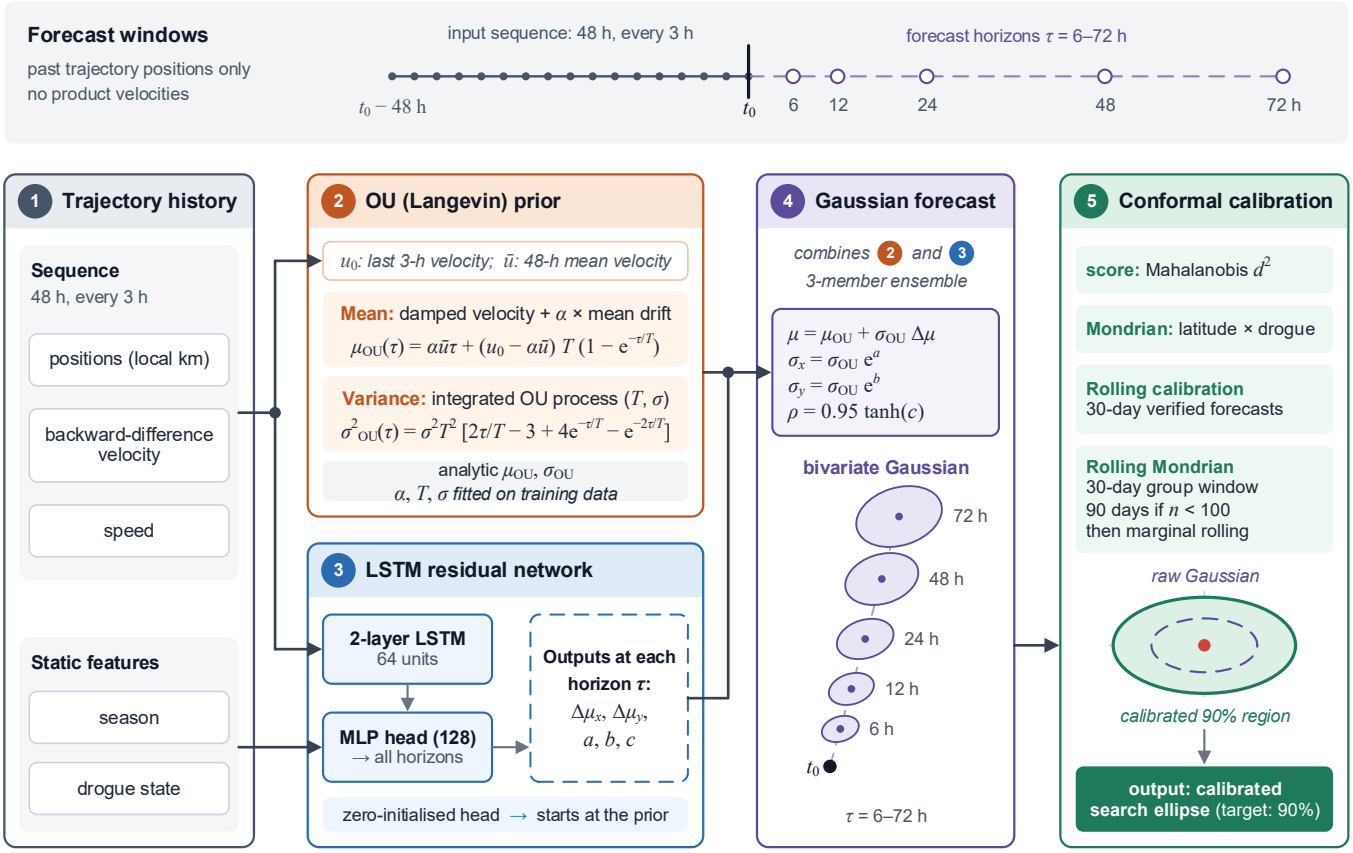

Forecast windows
past trajectory positions only
no product velocities
input sequence: 48 h, every 3 h
forecast horizons τ = 6–72 h
t0 − 48 h
t0
6
12
24
48
72 h
1 Trajectory history
Sequence
48 h, every 3 h
positions (local km)
backward-difference velocity
speed
Static features
season
drogue state
2 OU (Langevin) prior
u0: last 3-h velocity; ū: 48-h mean velocity
Mean: damped velocity + α × mean drift
μOU(τ) = αūτ + (u0 − αū) T (1 − e^{−τ/T})
Variance: integrated OU process (T, σ)
σ²OU(τ) = σ²T² [2τ/T − 3 + 4e^{−τ/T} − e^{−2τ/T}]
analytic μOU, σOU
α, T, σ fitted on training data
3 LSTM residual network
2-layer LSTM
64 units
MLP head (128)
→ all horizons
Outputs at each horizon τ:
Δμx, Δμy,
a, b, c
zero-initialised head → starts at the prior
4 Gaussian forecast
combines 2 and 3
3-member ensemble
μ = μOU + σOU Δμ
σx = σOU e^a
σy = σOU e^b
ρ = 0.95 tanh(c)
bivariate Gaussian
72 h
48 h
24 h
12 h
6 h
t0
τ = 6–72 h
5 Conformal calibration
score: Mahalanobis d²
Mondrian: latitude × drogue
Rolling calibration
30-day verified forecasts
Rolling Mondrian
30-day group window
90 days if n < 100
then marginal rolling
raw Gaussian
calibrated 90% region
output: calibrated search ellipse (target: 90%)